STRATHMORE UNIVERSITY • iLabAfrica • DSA 8401 Capstone Proposal

Kenya Land Readiness Atlas: Turning Satellite Pixels into Land Decisions

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1. Problem Statement

A county planner in Kisumu must decide where 3,000 affordable homes should go. Which land is farmable, which floods, and which is filling with buildings? The answer arrives months late, as a raw raster that only a GIS specialist can read. **The gap is not a lack of satellites; it is a lack of decisions built on them.**

I will build a system that closes it. From 10-band Sentinel-2 patches, it will classify land into **six classes** and return a **Settle** / **Farm** / **Protect** recommendation with a flood-exposure flag and a plain-language reason. The hard part is geographic transfer: a model that works near Kisumu can fail in Marsabit. Can a model trained on some Kenyan climate zones classify an unseen zone reliably, with calibrated confidence?

2. Data Source

Sentinel-2 surface reflectance (10 bands) and SRTM elevation come from Google Earth Engine, with labels from ESA WorldCover v200. My 8,356-point baseline took about 5 hours, so I will scale to 25,000 points as 64×64 patches across four zones: Arid (Marsabit), Coast (Kilifi), Highland (Nyeri/Kirinyaga) and Lake Basin (Kisumu).

Quality issues. **WorldCover is modelled, not ground truth**, so labels are noisy; accuracy is model-against-model until I hand-label 300 to 500 points. Classes are imbalanced (22 Arid water points).

3. Planned Approach

Baseline. LightGBM on eight spectral indices scores 0.648 macro-F1 under block CV but only 0.415 under LORO (leave-one-region-out), falling to 0.177 in the Arid zone, where dry soil is mistaken for built-up land.

Model. I will fine-tune EfficientNet-B0 with **Spectral Channel Attention**, focal loss and temperature scaling, targeting LORO macro-F1 above 0.60 and calibration error (ECE) at or below 0.08. Evaluation uses **buffered spatial block CV** and LORO, with a zero-leakage test.

Why these techniques. A CNN sees patch texture that tabular models cannot. I exclude random cross-validation (it overstates spatial performance), pixel-level segmentation (my labels are sampled points) and LLM fine-tuning (planning documents change, and RAG can cite them).

Serving. A scripted pipeline reproduces from scratch. FastAPI serves the CPU model through Docker Compose, with GitHub Actions tests and Evidently AI drift monitoring.

Decision support for the planner (RAG). The classifier says what the land is; planners also need what the rules allow. On the Streamlit map, a planner clicks a site and asks, “Can this land host affordable housing?” The **RAG copilot** retrieves passages from county planning documents (FAISS), reads the model’s class, confidence and flood flag, and returns a short answer with **cited sources and an uncertainty note**. It answers only from retrieved text and says when evidence is missing. Its outputs feed the **site-readiness layer** of my allocation framework as evidence only, never beneficiary eligibility. The planner decides.

